

## Assingment-1

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### Task (B) – Test Case Design Sheet

#### Constructor Test Cases

| TC ID | Method / Feature                                | Preconditions | Inputs                                                                                                     | Expected Result                                             | Type     |
|-------|-------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------|
| TC-01 | Constructor – valid                             | None          | productId="ITEM-99",<br>location="LOC-A1",<br>initialOnHand=50,<br>reorderThreshold=10,<br>maxCapacity=100 | Object created<br>successfully, reserved=0,<br>available=50 | Normal   |
| TC-02 | Constructor – null<br>productId                 | None          | productId=null                                                                                             | IllegalArgumentException<br>is thrown                       | Error    |
| TC-03 | Constructor –<br>blank location                 | None          | location=" "                                                                                               | IllegalArgumentException<br>is thrown                       | Error    |
| TC-04 | Constructor –<br>initialOnHand ><br>maxCapacity | None          | initialOnHand=200,<br>maxCapacity=100                                                                      | IllegalArgumentException<br>is thrown                       | Boundary |

### changeLocation Test Cases

| TC ID | Method / Feature       | Preconditions | Inputs               | Expected Result                    | Type   |
|-------|------------------------|---------------|----------------------|------------------------------------|--------|
| TC-05 | changeLocation – valid | Stock exists  | newLocation="LOC-B2" | Location is updated successfully   | Normal |
| TC-06 | changeLocation – null  | Stock exists  | newLocation=null     | IllegalArgumentException is thrown | Error  |
| TC-07 | changeLocation – blank | Stock exists  | newLocation=" "      | IllegalArgumentException is thrown | Error  |

### addStock Test Cases

| TC ID | Method / Feature            | Preconditions              | Inputs    | Expected Result                    | Type        |
|-------|-----------------------------|----------------------------|-----------|------------------------------------|-------------|
| TC-08 | addStock – valid            | onHand=50, maxCapacity=100 | amount=25 | onHand=75, available=75            | Normal      |
| TC-09 | addStock – zero             | onHand=50                  | amount=0  | IllegalArgumentException is thrown | Error       |
| TC-10 | addStock – exceeds capacity | onHand=50, maxCapacity=100 | amount=60 | IllegalStateException is thrown    | Boundary    |
| TC-11 | addStock – performance      | onHand=50                  | amount=10 | Method completes within 1 second   | Performance |

### removeDamaged Test Cases

| TC ID | Method / Feature               | Preconditions | Inputs     | Expected Result                    | Type     |
|-------|--------------------------------|---------------|------------|------------------------------------|----------|
| TC-12 | removeDamaged – valid          | onHand=50     | amount=15  | onHand=35                          | Normal   |
| TC-13 | removeDamaged – negative       | onHand=50     | amount=-3  | IllegalArgumentException is thrown | Error    |
| TC-14 | removeDamaged – exceeds onHand | onHand=50     | amount=500 | IllegalStateException is thrown    | Boundary |

### Reserve Test Cases

| TC ID | Method / Feature            | Preconditions | Inputs     | Expected Result                    | Type     |
|-------|-----------------------------|---------------|------------|------------------------------------|----------|
| TC-15 | reserve – valid             | available=50  | amount=20  | reserved=20, available=30          | Normal   |
| TC-16 | reserve – exceeds available | available=50  | amount=500 | IllegalStateException is thrown    | Boundary |
| TC-17 | reserve – zero              | available=50  | amount=0   | IllegalArgumentException is thrown | Error    |

### releaseReservation Test Cases

| TC ID | Method / Feature                      | Preconditions | Inputs     | Expected Result                 | Type     |
|-------|---------------------------------------|---------------|------------|---------------------------------|----------|
| TC-18 | releaseReservation – valid            | reserved=25   | amount=10  | reserved=15                     | Normal   |
| TC-19 | releaseReservation – exceeds reserved | reserved=10   | amount=100 | IllegalStateException is thrown | Boundary |

### shipReserved Test Cases

| TC ID | Method / Feature       | Preconditions             | Inputs     | Expected Result                 | Type     |
|-------|------------------------|---------------------------|------------|---------------------------------|----------|
| TC-20 | shipReserved – valid   | onHand=50,<br>reserved=15 | amount=10  | onHand=40, reserved=5           | Normal   |
| TC-21 | shipReserved – invalid | reserved=12               | amount=400 | IllegalStateException is thrown | Boundary |

### Reorder Logic Test Cases

| TC ID | Method / Feature        | Preconditions                 | Inputs | Expected Result | Type     |
|-------|-------------------------|-------------------------------|--------|-----------------|----------|
| TC-22 | isReorderNeeded – false | available=50,<br>threshold=10 | —      | returns false   | Decision |
| TC-23 | isReorderNeeded – true  | available=5, threshold=10     | —      | returns true    | Boundary |

### toString and getter Test Cases

| TC ID | Method / Feature           | Preconditions              | Inputs | Expected Result                                                                        | Type   |
|-------|----------------------------|----------------------------|--------|----------------------------------------------------------------------------------------|--------|
| TC-24 | toString – standard output | ProductStock object exists | —      | Returned string contains productId, location, onHand, reserved, available, maxCapacity | Normal |
| TC-25 | getAvailable               | onHand=50,<br>reserved=20  | —      | Returns 30                                                                             | Normal |
| TC-26 | getMaxCapacity             | ProductStock exists        | —      | Returns correct maxCapacity                                                            | Normal |

## JUnit Execution & Coverage Evidence

The screenshot displays the IntelliJ IDEA interface during a JUnit test run. The top toolbar shows the 'Run' button (a green play icon) and the 'Coverage' button (a green icon with a checkmark). The 'Coverage' tab is active, showing the test results for 'ProductStockSuite (org.osaid)'. The test suite consists of 25 tests, all of which passed, with a total execution time of 123 ms. The test results are listed in a table with columns for the test name, status, and duration. The tests include 'Test.ProductStock - Complete inventory', 'addStock increases onHand without', 'Constructor initializes attributes correctly', 'removeMaxCapacity supports increase', 'updateReorderThreshold accepts valid', 'toString should include key fields', 'changeLocation rejects null or blank', 'releaseReservation decreases reserved', 'addStock completes quickly under normal', 'changeLocation updates the internal', 'addStock rejects zero or negative input', 'reserve transfers quantity from available', 'Backorder computation (feature under evaluation)', 'releaseReservation rejects invalid value', 'removeDamaged reduces onHand availability', 'reserve rejects zero, negative, or above', 'addStock does not allow exceeding', 'updateMaxCapacity rejects invalid capacity', 'updateReorderThreshold rejects negative', 'Constructor rejects negative or incorrect', 'removeDamaged rejects values greater', 'shipReserved rejects zero, negative, or incorrect', 'shipReserved correctly reduces both', 'Constructor rejects invalid product ID', 'Reorder behavior tests', 'Reorder NOT needed when available', and 'Reorder needed when available'. The test results are followed by a detailed log of the test execution, showing the state of the 'ProductStock' object at various points in the test suite. The log includes the following information: 'Test.ProductStock Test Suite Started', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=75, reserved=0, available=75, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=0, available=50, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=0, available=50, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=0, available=50, reorderThreshold=120, maxCapacity=150)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=0, available=50, reorderThreshold=100, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=0, available=50, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=15, available=35, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=60, reserved=0, available=60, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-B2', onHand=50, reserved=0, available=50, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=0, available=50, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock(productId='ITEM-99', location='LOC-A1', onHand=50, reserved=20, available=30, reorderThreshold=10, maxCapacity=100)', 'Test.ProductStock Test Suite Completed', and 'Process finished with exit code 0'.

The screenshot displays the IntelliJ IDEA interface showing the Project view on the left and the Coverage view on the right. The Project view shows the project structure, including the 'src' directory, 'main' package, 'java' package, 'org.osaid' package, 'resources' directory, 'test' directory, 'java' package, 'org.osaid' package, 'ProductStockSuite' class, 'ProductStockTest' class, 'target' directory, 'gitignore' file, 'pom.xml' file, 'External Libraries' section, and 'Scratches and Consoles' section. The Coverage view shows the coverage for 'ProductStockSuite'. The coverage table is as follows:

| Element      | Class, %   | Method, %    | Line, %     | Branch, %   |
|--------------|------------|--------------|-------------|-------------|
| org.osaid    | 50% (1/2)  | 94% (18/19)  | 92% (71/77) | 89% (52/58) |
| Main         | 0% (0/1)   | 0% (0/1)     | 0% (0/3)    | 0% (0/2)    |
| ProductStock | 100% (1/1) | 100% (18/18) | 95% (71/74) | 92% (52/56) |

Coverage Summary

| Element             | Class Coverage | Method Coverage | Line Coverage | Branch Coverage |
|---------------------|----------------|-----------------|---------------|-----------------|
| org.osaid (overall) | 50% (1/2)      | 94% (18/19)     | 92% (71/77)   | 89% (52/58)     |
| ProductStock        | 100% (1/1)     | 100% (18/18)    | 95% (71/74)   | 92% (52/56)     |
| Main                | 0% (0/1)       | 0% (0/1)        | 0% (0/3)      | 0% (0/2)        |

Conclusion

الاختبارات التي تم تنفيذها بتغطي أغلب شغل كلاس **ProductStock** ونسبة عالية وجود 100% تغطية ميثودز وأكثر من 90% تغطية أسطر وفروع بيعطي ثقة كبيرة إنو الكود صحيح ومجرب منيح

Best Wishes